

05_Class_Activity

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In-Class Activity 5: Probability and Statistical Inference

What did we do last time?

In our previous activity, we:

- Created and interpreted frequency distributions (histograms)
- Compared data between groups using side-by-side histograms
- Explored how sample size affects our understanding of populations
- Created density plots and calculated probabilities

Today's focus:

Today we'll focus on:

- t-distribution and when to use it
- Calculating and interpreting standard error
- Creating confidence intervals
- Conducting **one-sample and two-sample t-tests**
- **Understanding statistical assumptions and their importance**

Setup

First, let's load the packages and data we'll be using:

```
# Load required packages
library(readxl)
library(tidyverse)
library(patchwork)
library(car) # For diagnostic tests

pine_df <- read_excel("data/class_pine needle length.xlsx")
pine_switch_df <- read_excel("data/class_pine needle length switched.xlsx")

head(pine_df)
```

```
# A tibble: 6 × 5
  group tree_no tree_char side length_mm
<chr>   <dbl> <chr>    <chr>    <dbl>
1 five     1 tree_1 sunny     22.7
2 five     1 tree_1 sunny     21.1
3 five     1 tree_1 sunny     18.6
4 five     1 tree_1 sunny     18.6
5 five     1 tree_1 sunny     21.0
6 five     1 tree_1 sunny     18.9
```

An issue we have is pseudoreplication

What is our sample unit? branches? needles? trees? sides?

- We need to get the averages of the data so it is no pseudoreplicated!
- We will group data by group, tree, tree_char, side and take the averages and save the average as length_mm
- We will also need to make dataframes of sunny and shady side of the data

```
# Now we need to average the sunny and shady sides as these are all pseudoreplicated
p_df <- pine_df %>%
  group_by(group, tree_no, tree_char, side) %>%
  summarise(length_mm = mean(length_mm, na.rm=TRUE))
```

`summarise()` has grouped output by 'group', 'tree_no', 'tree_char'. You can override using the `.groups` argument.

```
ps_df <- pine_switch_df %>%
  group_by(group, tree_no, tree_char, side) %>%
  summarise(length_mm = mean(length_mm, na.rm=TRUE))
```

`summarise()` has grouped output by 'group', 'tree_no', 'tree_char'. You can override using the `.groups` argument.

```
ps_shady_df <- ps_df %>%
  filter(side == "shady")

ps_sunny_df <- ps_df %>%
  filter(side == "sunny")

head(ps_shady_df) %>% arrange(tree_no)
```

```
# A tibble: 6 × 5
# Groups:   group, tree_no, tree_char [6]
  group          tree_no tree_char side length_mm
<chr>          <dbl> <chr>   <chr>   <dbl>
1 five              1 tree_1  shady    20.3
2 big_fat_fecund_female_fish  2 tree_2  shady    15.4
3 bill              3 tree_3  shady    16.7
4 ciabatta          5 tree_5  shady    19.1
5 moose_walkin      7 tree_7  shady    20.7
6 fake_data        8 tree_8  shady    17.4
```

Part 1: Exploring the Data

Before conducting statistical tests, it's important to understand your data.

💡 Practice Exercise 1: Summary data

Let's create summary data of needle lengths from each side to visualize their distributions.

```
stats_df <- ps_df %>%  
  group_by(side) %>%  
  summarize(  
    mean_length = mean(length_mm, na.rm = TRUE),  
    sd_length = sd(length_mm, na.rm = TRUE),  
    se_length = sd(length_mm, na.rm = TRUE) / sum(!is.na(length_mm))^.5,  
    count = sum(!is.na(length_mm)),  
    .groups = "drop"  
  )  
stats_df
```

```
# A tibble: 2 × 5  
  side mean_length sd_length se_length count  
  <chr>      <dbl>      <dbl>      <dbl> <int>  
1 shady      17.6         2.51       0.886     8  
2 sunny      16.2         2.64       0.934     8
```

```
# CAN YOU THINK OF AN EASIER WAY?
```

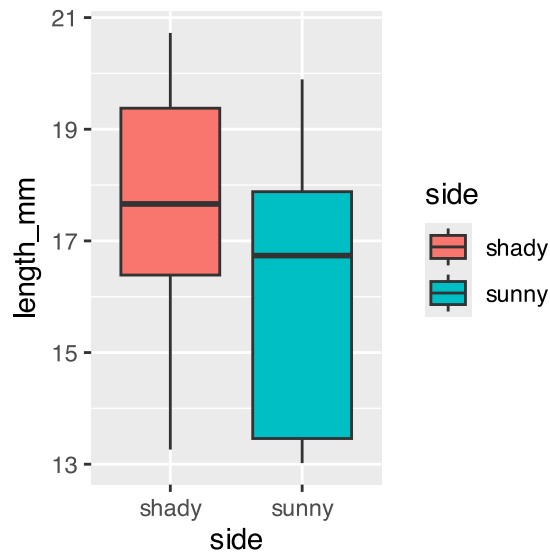
Part 1: Exploring the Data

Before conducting statistical tests, it's important to understand your data.

💡 Practice Exercise 1: Creating Histograms

Let's create histograms of needle lengths from each lake to visualize their distributions.

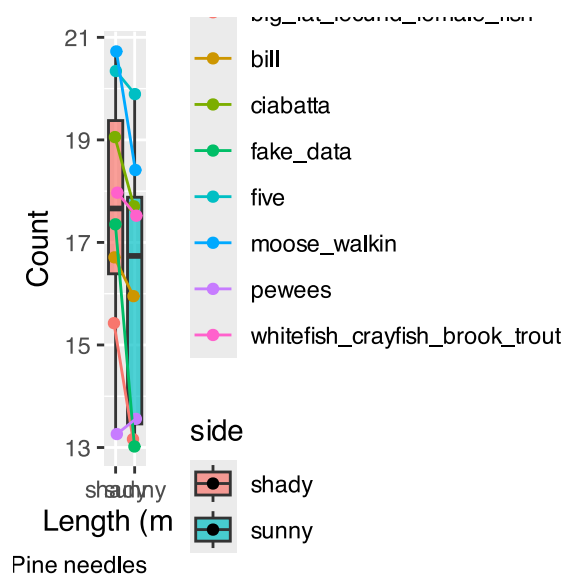
```
needle_box_plot <- ps_df %>%  
  ggplot(aes(x=side, y=length_mm, fill = side))+  
  geom_boxplot()  
needle_box_plot
```



Part 2: Exploring data a different way

We want to see variation in a different way and how groups measured data on sunny and shady sides

```
pd <- position_dodge2(width = 0.2)  
  
p_plot <- ps_df %>%  
  ggplot(aes(side, length_mm, fill=side)) +  
  geom_boxplot(alpha = 0.7) +  
  geom_point(aes(group = group, color= group),  
             position = pd)+  
  geom_line(aes(group = group, color= group),  
            position = pd)+  
  labs(x = "Length (mm)", y = "Count", caption = "Pine needles")  
p_plot
```



Part 3: Testing Assumptions

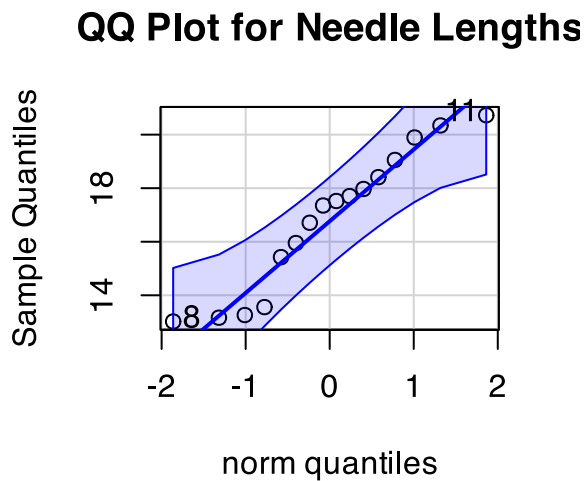
Before conducting a t-test, we need to check if our data meets the necessary assumptions:

1. **Normality:** The data should be approximately normally distributed
2. **Independence:** Observations should be independent
3. **No extreme outliers:** Outliers can heavily influence t-test results

Let's check the normality assumption for shady needle lengths:

💡 Practice Exercise 2: Checking Normality of all data but does not do by group!

```
# Create a QQ plot to check normality
# QQ plots compare our data to a theoretical normal distribution
# Points should roughly follow the line if data is normally distributed
qqPlot(ps_df$length_mm,
       main = "QQ Plot for Needle Lengths",
       ylab = "Sample Quantiles")
```



```
[1] 8 11
```

Shapiro-Wilk test on all data

What do we look at here and how do we interpret?

Hoping for non significant!!!

```
# Also perform a formal test of normality using the Shapiro-Wilk test
# Null hypothesis: Data is normally distributed
# If  $p > 0.05$ , we don't reject the assumption of normality
shapiro_test <- shapiro.test(ps_df$length_mm)
print(shapiro_test)
```

Shapiro-Wilk normality test

```
data: ps_df$length_mm
W = 0.92754, p-value = 0.2228
```

QQ Plot for Sunny Data

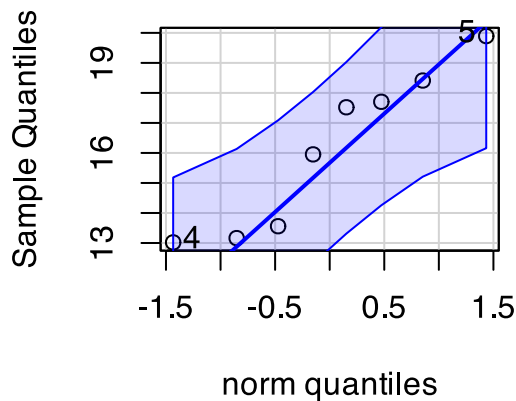
💡 Practice Exercise 8: Test normality of pine needle lengths on sunny side

qqplots

Note you need to test each groups separately...

```
# QQ Plot for windward group
qqPlot(ps_sunny_df$length_mm,
       main = "QQ Plot for sunny needle length",
       ylab = "Sample Quantiles")
```

QQ Plot for sunny needle leng



```
[1] 5 4
```

Shapiro-Wilk Test for Sunny data

💡 Practice Exercise 9: Test normality of I8 lengths

Shapiro-Wilk test

Note you need to test each groups separately...

```
# Shapiro-Wilk test for windward group
shapiro.test(ps_sunny_df$length_mm)
```

Shapiro-Wilk normality test

```
data: ps_sunny_df$length_mm
W = 0.89994, p-value = 0.2886
```

QQ Plot for Shady data

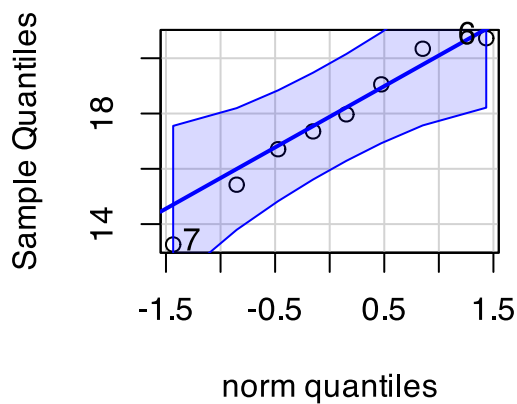
💡 Practice Exercise 10: Test normality of shady lengths

qqplots

Note you need to test each groups separately...

```
# You can also test the i3  
# QQ Plot for leeward group  
qqPlot(ps_shady_df$length_mm,  
        main = "QQ Plot for shady needle length",  
        ylab = "Sample Quantiles")
```

QQ Plot for shady needle leng



```
[1] 7 6
```

Shapiro-Wilk for Shady data

💡 Practice Exercise 11: Test normality of shady lengths

Shapiro-Wilk test

Note you need to test each groups separately...

```
# Shapiro-Wilk test for leeward group  
shapiro.test(ps_shady_df$length_mm)
```

Shapiro-Wilk normality test

```
data: ps_shady_df$length_mm  
W = 0.96639, p-value = 0.8683
```

Combined Normality Test

Note you can do this a lot easier with piping of the data...

💡 Practice Exercise 12: Test Normality at one time

There are always a lot of ways to do this in R

```
# there are always two ways
# Test for normality using Shapiro-Wilk test for each wind group
# All in one pipeline using tidyverse approach
normality_results <- ps_df %>%
  group_by(side) %>%
  summarize(
    shapiro_stat = shapiro.test(length_mm)$statistic,
    shapiro_p_value = shapiro.test(length_mm)$p.value,
    normal_distribution = if_else(shapiro_p_value > 0.05, "Normal", "Non-normal")
    # above we are using an ifelse test which is a great oneliner
  )

# Print the results
print(normality_results)
```

```
# A tibble: 2 × 4
  side shapiro_stat shapiro_p_value normal_distribution
<chr>      <dbl>          <dbl> <chr>
1 shady      0.966            0.868 Normal
2 sunny      0.900            0.289 Normal
```

Test for Equal Variances of Sunny and Shady

💡 Practice Exercise 13: Test equal variances

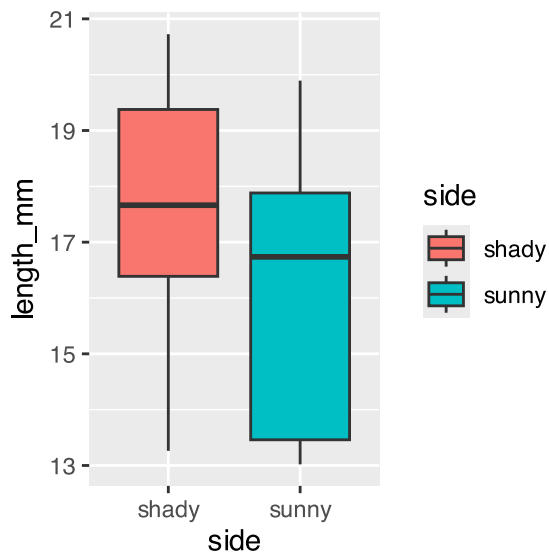
Levene's test can be done on the original dataframe

```
# Method 1: Using car package's leveneTest
# This is often preferred as it's more robust to departures from normality
levene_result <- leveneTest(length_mm ~ side, data = ps_df)
levene_result
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.2062 0.6567
      14
```

Check for outliers

```
# Check for outliers using a boxplot
needle_box_plot
```



💡 Tip

How to interpret these results:

- The QQ plot: Points should follow the straight line if data is normally distributed
- Shapiro-Wilk test: If $p > 0.05$, we don't reject the assumption of normality
- Boxplot: Look for points beyond the whiskers as potential outliers

Part 4: One-Sample t-Test

A one-sample t-test compares a sample mean to a specific value.

Let's test if the mean needle length on the shady side is 15 or not?

💡 Practice Exercise: One-Sample t-Test

```
# Calculate the mean of shady needles?
shady_mean <- mean(ps_shady_df$length_mm, na.rm=TRUE)
shady_mean
```

```
[1] 17.60561
```

```
# what is the mean
# 17.60561

ps_shade_mean <- mean(ps_shady_df$length_mm, na.rm = TRUE)
cat("Mean:", round(ps_shade_mean, 1), "mm\n")
```

```
Mean: 17.6 mm
```

```
# Perform a one-sample t-test
t_test_result <- t.test(ps_shady_df$length_mm, mu = 15)

# View the test results
t_test_result
```

One Sample t-test

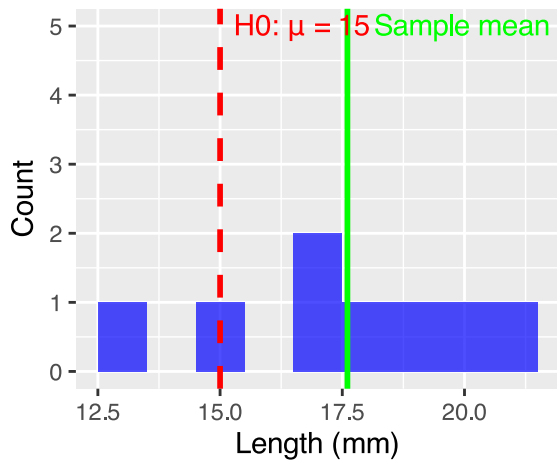
```
data: ps_shady_df$length_mm
t = 2.9414, df = 7, p-value = 0.02167
alternative hypothesis: true mean is not equal to 15
95 percent confidence interval:
 15.51092 19.70030
sample estimates:
mean of x
 17.60561
```

```
# Create a visualization of the test
ps_shady_df %>%
  ggplot(aes(x = length_mm)) +
  geom_histogram(binwidth = 1, fill = "blue", alpha = 0.7) +
  geom_vline(xintercept = 15, color = "red", linetype = "dashed", linewidth = 1) +
  geom_vline(xintercept = shady_mean, color = "green", linewidth = 1) +
  annotate("text", x = 15, y = 5, label = "H0:  $\mu = 15$ ", color = "red", hjust = -0.1) +
  annotate("text", x = ps_shade_mean, y = 5,
    label = paste("Sample mean =", round(ps_shade_mean, 1)),
    color = "green", hjust = -0.1) +
  labs(
    title = "One-Sample t-Test: Shady Needle Lengths",
    subtitle = paste(
      "t =", round(t_test_result$statistic, 2),
      ", p =", format.pval(t_test_result$p.value, digits = 3)),
```

```
x = "Length (mm)",
y = "Count")
```

One-Sample t-Test: Shady Neece

$t = 2.94$, $p = 0.0217$



Tip

Interpret the results:

1. What was the null hypothesis? $H_0: \mu = 15\text{mm}$
2. What was the alternative hypothesis? $H_1: \mu \neq 15\text{mm}$
3. What does the p-value tell us? (Is $p < 0.05$?)
4. Should we reject or fail to reject the null hypothesis?
5. What is the practical interpretation for biologists?

Part 5: Confidence Intervals

A confidence interval gives us a range of plausible values for the population mean.

For a 95% confidence interval using the t-distribution:

$$95\% \text{ CI} = \bar{x} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$$

Where:

- $\bar{x}$ is the sample mean
- s is the sample standard deviation
- n is the sample size
- $t_{\alpha/2, n-1}$ is the critical t-value with $n-1$ degrees of freedom

💡 Practice Exercise 4: Calculating Confidence Intervals

Let's calculate the 95% confidence interval for shady needle lengths:

```
# Extract sample statistics
shady_stats <- ps_shady_df
shady_mean <- mean(ps_shady_df$length_mm, na.rm=TRUE)
shady_se <- sd(ps_shady_df$length_mm, na.rm=TRUE)/sum(!is.na(ps_shady_df$length_mm))^.5
shady_n <- sum(!is.na(ps_shady_df$length_mm))

# Find the critical t-value for 95% confidence with n-1 degrees of freedom
# qt(0.975, df) gives the t-value for a 95% confidence interval (two-tailed)
t_critical <- qt(0.975, df = shady_n - 1)
cat("Critical t-value for", shady_n-1, "degrees of freedom:", round(t_critical, 3), "\n")
```

Critical t-value for 7 degrees of freedom: 2.365

```
# Calculate the confidence interval
shady_ci_lower <- shady_mean - t_critical * shady_se
shady_ci_upper <- shady_mean + t_critical * shady_se

# Display the confidence interval
cat("95% Confidence Interval for Shady needle mean length:",
    round(shady_ci_lower, 1), "to", round(shady_ci_upper, 1), "mm\n")
```

95% Confidence Interval for Shady needle mean length: 15.5 to 19.7 mm

```
# Compare this to a confidence interval using the normal approximation (z = 1.96)
z_ci_lower <- shady_mean - 1.96 * shady_se
z_ci_upper <- shady_mean + 1.96 * shady_se

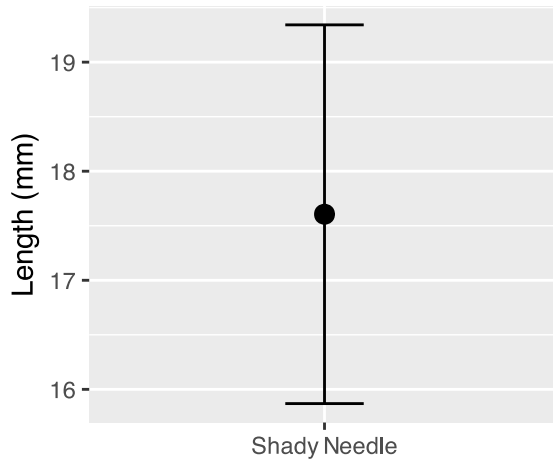
cat("95% CI using normal approximation:",
    round(z_ci_lower, 1), "to", round(z_ci_upper, 1), "mm\n")
```

95% CI using normal approximation: 15.9 to 19.3 mm

```
# Visualize the confidence interval
ggplot() +
  geom_errorbar(aes(x = "Shady Needle",
                    ymin = z_ci_lower,
                    ymax = z_ci_upper),
                width = 0.2) +
  geom_point(aes(x = "Shady Needle", y = shady_mean), size = 3) +
  labs(title = "Mean Needle Length with 95% Confidence Interval",
       subtitle = "Shady Needle",
       x = NULL,
       y = "Length (mm)")
```

Mean Needle Length with 95%

Shady Needle



💡 Interpretation:

- We are 95% confident that the true population mean needle length on shady side falls within this interval
- Note the small difference between using the t-distribution vs. normal approximation

Part 6: Two-Sample t-Test

A two-sample t-test compares means from two independent groups.

Let's compare needle lengths between sunny and shady sides

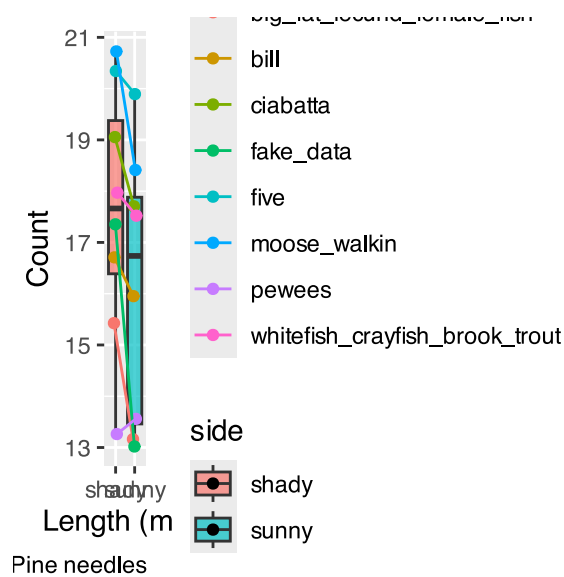
```
# Summarize pine needle data by wind exposure
p_summary <- ps_df %>%
  group_by(side) %>%
  summarize(
    mean_length = mean(length_mm),
    sd_length = sd(length_mm),
    n = sum(!is.na(length_mm)),
    se_length = sd_length / sqrt(n)
  )

# Display the summary statistics
print(p_summary)
```

```
# A tibble: 2 × 5
  side mean_length sd_length    n se_length
<chr>    <dbl>    <dbl> <int>    <dbl>
1 shady     17.6     2.51     8     0.886
2 sunny     16.2     2.64     8     0.934
```

Look at the plot of needle lengths

```
# Create a boxplot to visualize the data
p_plot
```



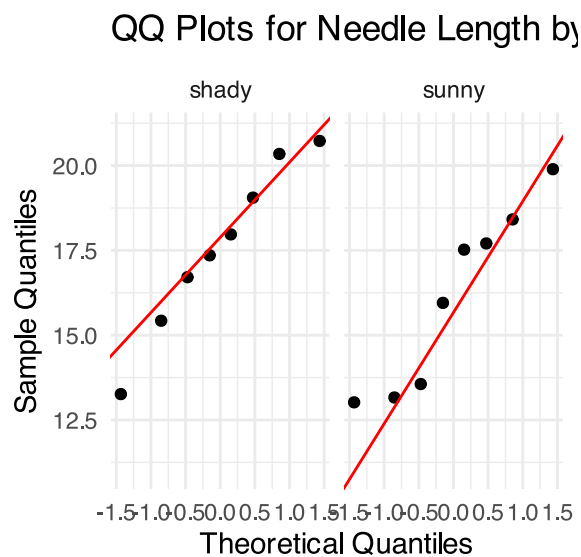
Before conducting the t-test, we should check the assumptions:

💡 Practice Exercise 5: Check Assumptions for Two-Sample t-Test

Normality with qq plot

```
# 1. Check for normality in each group using QQ plots

ggplot(ps_df, aes(sample = length_mm)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  facet_wrap(~ side) +
  labs(title = "QQ Plots for Needle Length by Side",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```



Normality with Shapiro-Wilk

```
# there are always two ways
# Test for normality using Shapiro-Wilk test for each wind group
# All in one pipeline using tidyverse approach
normality_results <- ps_df %>%
  group_by(side) %>%
  summarize(
    shapiro_stat = shapiro.test(length_mm)$statistic,
    shapiro_p_value = shapiro.test(length_mm)$p.value,
    normal_distribution = if_else(shapiro_p_value > 0.05, "Normal", "Non-normal")
    # above we are using an ifelse test which is a great oneliner
  )

# Print the results
normality_results
```

```
# A tibble: 2 × 4
  side shapiro_stat shapiro_p_value normal_distribution
<chr>    <dbl>         <dbl> <chr>
1 shady     0.966         0.868 Normal
2 sunny     0.900         0.289 Normal
```

Homogeneity of variance

Tip

```
# 2. Check for equal variances using Levene's test
# H0: Variances are equal
# H1: Variances are not equal
levene_result <- leveneTest(length_mm ~ side, data = ps_df)
```

```
Warning in leveneTest.default(y = y, group = group, ...): group coerced to
factor.
```

```
levene_result
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.2062 0.6567
      14
```

Tip

Interpreting the assumption checks:

- QQ plots: Do points approximately follow the line for both groups?
- Shapiro - Wilk: is the outcome $p > 0.05$? Then don't reject H_0 that they are normally distributed
- Levene's test: If $p > 0.05$, we don't reject the assumption of equal variances

Now let's conduct the two-sample t-test:

💡 Practice Exercise 6: Two-Sample t-Test

Calculate the T Test

```
# Perform a two-sample t-test
# H0:  $\mu_1 = \mu_2$  (The mean needle lengths are equal)
# H1:  $\mu_1 \neq \mu_2$  (The mean needle lengths are different)

# var.equal=TRUE uses the standard t-test (pooled variance)
# var.equal=FALSE uses Welch's t-test (for unequal variances)
t_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = TRUE)

# Display the test results
t_test_result
```

Two Sample t-test

```
data: length_mm by side
t = 1.1279, df = 14, p-value = 0.2783
alternative hypothesis: true difference in means between group shady and group sunny is
not equal to 0
95 percent confidence interval:
 -1.309330  4.214005
sample estimates:
mean in group shady mean in group sunny
      17.60561      16.15328
```

Compare to a Welch's Two Sample T Test when variance is unequal

```
# Perform a two-sample t-test
# H0:  $\mu_1 = \mu_2$  (The mean needle lengths are equal)
# H1:  $\mu_1 \neq \mu_2$  (The mean needle lengths are different)

# var.equal=TRUE uses the standard t-test (pooled variance)
# var.equal=FALSE uses Welch's t-test (for unequal variances)
welch_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = FALSE)

# Display the test results
welch_test_result
```

Welch Two Sample t-test

```
data: length_mm by side
t = 1.1279, df = 13.96, p-value = 0.2784
alternative hypothesis: true difference in means between group shady and group sunny is not
equal to 0
95 percent confidence interval:
 -1.310069  4.214743
sample estimates:
mean in group shady mean in group sunny
      17.60561      16.15328
```

Compare to a Paired T Test

```
# Perform a two-sample t-test
# H0:  $\mu_1 = \mu_2$  (The mean needle lengths are equal)
# H1:  $\mu_1 \neq \mu_2$  (The mean needle lengths are different)

# var.equal=TRUE uses the standard t-test (pooled variance)
# var.equal=FALSE uses Welch's t-test (for unequal variances)
paired_test_result <- t.test(length_mm ~ side, data = ps_df, pair = TRUE)

# Display the test results
paired_test_result
```

Paired t-test

```
data: length_mm by side
t = 2.7818, df = 7, p-value = 0.02723
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.2178092 2.6868652
sample estimates:
mean difference
 1.452337
```

Look at Effect Size

💡 Tip

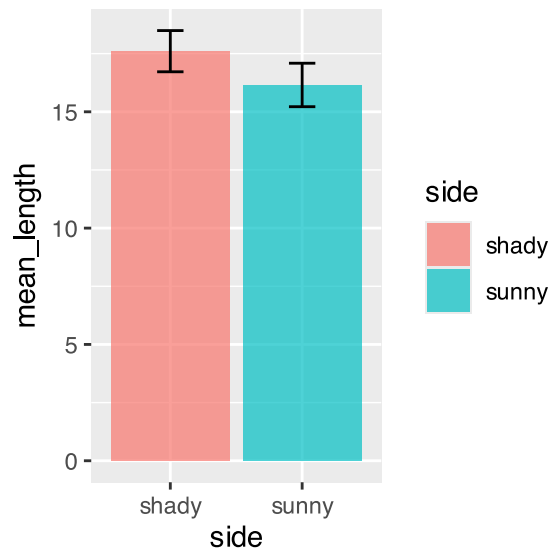
```
# Calculate the mean difference
stats_df %>%
  summarize(difference = mean_length[side == "shady"] - mean_length[side == "sunny"])
```

```
# A tibble: 1 × 1
  difference
    <dbl>
1       1.45
```

Visualize the plot

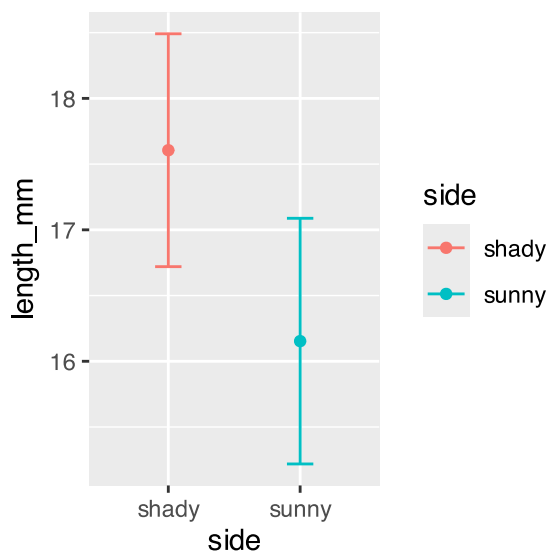
💡 Tip

```
# Visualize the results with a mean and error bar plot
ggplot(stats_df, aes(x = side, y = mean_length, fill = side)) +
  geom_bar(stat = "identity", alpha = 0.7) +
  geom_errorbar(aes(ymin = mean_length - se_length,
                    ymax = mean_length + se_length),
                width = 0.2)
```



Alternative plot method

```
ggplot(ps_df, aes(x = side, y = length_mm, color = side)) +
  stat_summary(fun=mean, geom="point")+
  stat_summary(fun.data = mean_se, geom="errorbar", width = 0.2)
```



Interpretation:



Tip

Interpret the results:

1. What was the null hypothesis?
2. What was the alternative hypothesis?
3. What does the p-value tell us?
4. Should we reject or fail to reject the null hypothesis?
5. What is the practical interpretation for botanists?

Part 8: Communicating Statistical Results

In scientific writing, it's important to report statistical results clearly and consistently.

Here's a standard format for reporting t-test results:

For a one-sample t-test: "A one-sample t-test showed that the mean needle length shady side ($M = [\text{mean}]$, $SD = [\text{sd}]$) was [significantly/not significantly] different from 15 mm, $t(df) = [\text{t-value}]$, $p = [\text{p-value}]$."

For a two-sample t-test: "A two-sample t-test revealed that pine needle lengths on the sunny side ($M = [\text{mean1}]$, $SD = [\text{sd1}]$) were [significantly/not significantly] [longer/shorter] than on the shady side ($M = [\text{mean2}]$, $SD = [\text{sd2}]$), $t(df) = [\text{t-value}]$, $p = [\text{p-value}]$."



Practice Exercise 8: Writing Statistical Results

Write properly formatted statements reporting the results of:

1. The one-sample t-test comparing mean shady needle length to 15mm
2. The two-sample t-test comparing pine needle lengths
3. The two-sample t-test comparing needle lengths between sides

Remember to include:

- Means and standard deviations for each group
- The t-value with degrees of freedom
- The p-value and whether the result is significant

Reflection Questions

1. How does the t-distribution differ from the normal distribution, and why does this matter for small samples?
2. What assumptions must be met to use a t-test, and what alternatives exist if these assumptions are violated?
3. What is the difference between statistical significance and practical importance?
4. How would the confidence interval change if we used a 99% confidence level instead of 95%?
5. How would you explain the concept of a p-value to someone with no statistical background?